Differentiation and Rates of Change

Section 1: Product and Quotient Rules

1 Differentiate $y = x^2 e^{3x}$ with respect to x.

a. $xe^{3x}(3x + 2)$

b. $2xe^{3x}$

c. $3x^2 e^{3x}$

d. $x^2 e^{3x} + 2x$

2 Find $\frac{dy}{dx}$ for $y = \frac{2x - 1}{x + 4}$.

a. $\frac{9}{(x + 4)^2}$

b. $\frac{2}{(x + 4)^2}$

c. $\frac{7}{(x + 4)^2}$

d. $\frac{2x + 9}{(x + 4)^2}$

3 Find the derivative of $f(x) = x \ln x$ evaluated at $x = e$.

a. 2

b. 1

c. e

d. 0

Section 2: Chain Rule and Composite Functions

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4 Differentiate $y = (3x^2 - 5)^4$ with respect to x.

a. $24x(3x^2 - 5)^3$

b. $12x(3x^2 - 5)^3$

c. $4(3x^2 - 5)^3$

d. $18x^2(3x^2 - 5)^3$

5 Find the derivative of $y = \sin^3(2x)$.

a. $6\sin^2(2x)\cos(2x)$

b. $3\sin^2(2x)\cos(2x)$

c. $6\sin(2x)\cos(2x)$

d. $-6\sin^2(2x)\cos(2x)$

6 Find the gradient of $y = \sqrt{4x + 9}$ at the point where $x = 4$.

a. $\frac{2}{5}$

b. $\frac{4}{5}$

c. $\frac{1}{5}$

d. 5

Section 3: Tangents, Normals and Stationary Points

7 Find the equation of the tangent to the curve $y = x^2 - 4x + 3$ at the point where $x = 3$.

a. $y = 2x - 6$

b. $y = 2x + 6$

c. $y = -2x$

d. $y = x - 3$

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8 Find the coordinates of the stationary point of $y = x^3 - 3x + 2$ in the first quadrant.

a. $(1, 0)$

b. $(1, 2)$

c. $(2, 4)$

d. $(0, 2)$

9 Determine the nature of the stationary point of $f(x) = 2x^3 - 9x^2 + 12x$ at $x = 2$.

a. Local Minimum

b. Local Maximum

c. Point of Inflexion

d. Vertical Asymptote

Section 4: Connected Rates of Change and Applied Optimization

10 The radius of a circular oil spill is increasing at 0.5 m/s. At what rate is the area increasing when the radius is 10 m?

a. 10π m^2/s

b. 20π m^2/s

c. 5π m^2/s

d. 100π m^2/s

11 A solid cylindrical metal rod of radius r and length h has fixed volume $V = 54\pi$. Express the surface area S in terms of r and find r minimizing S.

a. $r = 3$

b. $r = 2$

c. $r = 6$

d. $r = 4$

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12 If $y = 3x^2 - 4x$ and x increases at a rate of 2 units/s, find the rate of increase of y when $x = 5$.

a. 52 units/s

b. 26 units/s

c. 30 units/s

d. 60 units/s

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